

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

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COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I T Gnaneswari(192372183) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

T Gnaneswari

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

- i) Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concepts of responsive web design and its importance. Describe key techniques CSS, Grid can be used to design adaptive layouts.

Responsive Web Design :-

Definition :-

Responsive web Design (RWD) is a web design approach that makes a website automatically adjust its layout according to different screen sizes such as mobile phones, tablets and desktop computers.

Importance :-

- Provide better user experience on all devices.
- Eliminate horizontal scrolling.
- Improves website accessibility.
- Increase mobile friendliness.
- Helps improve SEO rankings.

CSS Techniques used :-

- Media queries - Apply different style for screen size
- flexible layouts - use Percentage (%) instead of fixed
- flexible Images - Images resize automatically.
- flexbox/ Grid - create responsive layouts easily.

using flexbox :-

flexbox automatically arranges items in rows or columns and adjusts spacing according of screen size

Sample code :-

```
<html>
<head>
<style>
  .container {
    display : flex;
    flex-wrap : wrap;
  }
  .box {
    flex : 1;
    padding : 20px;
    margin : 10px;
  }
  @media (max-width : 768px) {
    .container {
      flex-direction : column;
    }
  }
```

</style>

<body>

<div class = "container">

<div class = "box">Home</div>

</div>

</body>

</html>.

Improvements :-

- use responsive images.
- Increase button size for mobile users.
- Maintain readable font sizes.
- optimize page loading speed.
- use CSS Grid for complex layouts.

2) client-side validation :

Definition :-

client-side validation checks user inputs in the browser before sending data to the server.

Purpose :

- Prevent invalid data entry.
- Improve user experience.
- Reduce server workload.
- Display instant error messages.

JavaScript Event Handling

JavaScript uses events such as :-

- Submit
- Click
- input
- change

The submit event checks all fields before the form is submitted.

Regular Expressions :

Regular expressions (Regex) are pattern used to validate data.

Example :

- Email
- phone number
- Password

Sample code :-

```
<html>
<body>
<form onsubmit = "return validate()">
  Email :
  <input type = "text" id = "email"><br><br>
  phone :
  <input type = "text" id = "phone"><br><br>
  <input type = "submit">
</form>
<script>
function validate() {
let email = document.getElementById ("email").value;
let phone = document.getElementById ("phone").value;
let emailpattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
let phonepattern = /^[0-9]{10}$/;
if (!emailpattern.test(email)) {
  alert ("Invalid Email");
  return false;
}
if (!phonepattern.test(phone)) {
  alert ("Invalid phone number");
  return false;
}
alert (" form submitted");
return true;
}
</script>
</body>
</html>
```

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Improvement :-

- show error messages beside input fields.
- use HTML5 validation (required, email, pattern).
- perform server-side validation for security.
- provide clear user instructions.

3)

CSS Box Model

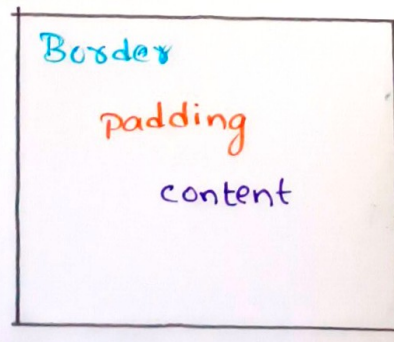
Definition :-

The CSS Box model describes how every HTML element is displayed as a rectangular box.

Components :-

1. Content
2. padding
3. Border
4. Margin.

Margin



CSS positioning :

- Static : default position.
- Relative : Move relative to original position.
- Absolute : positioned relative to nearest positioned parent.
- fixed : fixed to browser window.

- Sticky : Sticks while scrolling.

Causes of layout Breaking :-

- fixed width elements.
- large images.
- Missing media queries.
- Incorrect positioning.
- overflowing content.

Media Queries Example:

```
• Container {  
    display : flex ;  
}  
@media (max-width : 768px) {  
    • container {  
        flex-direction : column ;  
    }  
}
```

Best practices :

- use flexbox or Grid.
- use percentage widths.
- Apply media queries.
- use box-sizing : border-box.
- Test on different screen sizes.

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Document object Model (DOM)

Introduction :-

Modern website often need to update their content without refreshing the entire webpages. This is achieved using Document object model (DOM) and Javascript. The DOM allows JS to access and modify HTML elements dynamically, making web applications faster, more interactive, and user-friendly.

Definition:-

The Document object model (DOM) is a tree-like structure that represents all HTML elements as objects. JavaScripts uses the DOM to access and modify webpage content dynamically.

Role of DOM :

- update text
- change style.
- Add or remove elements.
- Handle user interactions.
- create dynamic webpages.

Javascript DOM manipulation

common methods: `getElementById()`

querySelector()

innerHTML

textContent

createElement()

appendChild()

Example :

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
  <h2 id="message">welcome</h2>
```

```
  <button onclick="changeText()">
```

```
    Click Here.
```

```
  </button>
```

```
<script>
```

```
  function changeText() {
```

```
    document.getElementById("message").innerHTML =
```

```
      "welcome to IP";
```

```
  }
```

```
</script>
```

```
</body>
```

```
</html>
```

Improvements :-

- update only required elements instead of whole page.
- use event listeners instead of inline events.
- Reduce unnecessary DOM operations.
- use cache for frequently accessed elements.
- Improve responsiveness using asynchronous updates.